THREE BRILLHART–LEHMER–SELFRIDGE PRIMALITY PROOFS FOR WAGSTAFF NUMBERS

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ABSTRACT. The Wagstaff numbers $W_p = (2^p+1)/3$ for odd primes p are the natural $+1$ companions of the Mersenne numbers. Known primality proofs for W_p with $p \geq 2617$ (as recorded in the Caldwell/Wagstaff-project archives [6, 5]) rely on the elliptic-curve primality proving algorithm of Atkin–Morain [2]; Chebyshev/Lucas-type tests, while available as compositeness criteria, remain conjectural on the sufficiency side. We present fully verified primality proofs of W_{2617} (788 digits), W_{10501} (3161 digits), and W_{12391} (3730 digits), independent of ECPP and relying only on classical $N-1$ machinery. The proofs apply the Brillhart–Lehmer–Selfridge (BLS) $N-1$ criterion [3] to the cyclotomic decomposition $2^{p-1}-1 = \prod_{d|p-1} \Phi_d(2)$, harvesting factored content from the Cunningham project tables [6] (used as evidence) and FactorDB [9] (used only as a discovery aid, with every retrieved factor re-certified). As an independent check on the $\mathbb{Z}[\sqrt{2}]$ arithmetic implementation, the Chua $N+1$ congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ — the $a = 3$ case of the Chua framework [7] with $\omega_3 = 3+2\sqrt{2}$, a necessary condition for Wagstaff primality — is verified at each W_p.

BLS $N-1$ requires $p-1$ sufficiently smooth that enough cyclotomic factors $\Phi_d(2)$ are fully factored. On the factorisation data consulted (Cunningham project tables and FactorDB, January–April 2026), $p = 10501$ and $p = 12391$ are the only exponents above 2617 in the known Wagstaff prime/PRP list meeting this ceiling (Remark 4.4). Every cofactor primality is certified unconditionally by APR-CL [1, 8]; the method is independent of the Chebyshev sufficiency conjecture, and every step is reproducible from the archived scripts.

1. INTRODUCTION

1.1. **Wagstaff numbers and the ECPP era.** For an odd prime p, the *Wagstaff number* is

$$W_p \;=\; \frac{2^p+1}{3}.$$

The exponents p for which W_p is prime form OEIS A000978; the Wagstaff primes themselves, as a sequence of integers, form OEIS A000979. All W_p with $p \leq 42737$ have been proved prime or composite (see Appendix B for the 29 proved exponents); 7 more probable primes are known up to the current computational horizon. In the Caldwell Prime Pages archive [5] and the Cunningham project tables [6] as consulted for this paper (January–April

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2026), the recorded primality proofs for Wagstaff numbers W_p with $p \geq 2617$ are ECPP proofs in the sense of Atkin–Morain [2]. We are not aware of previously published BLS $N-1$ proofs for W_{2617}, W_{10501}, or W_{12391}.

The Mersenne counterparts $M_p = 2^p - 1$ admit the Lucas–Lehmer test [10], a deterministic standalone primality criterion. No deterministic standalone test of comparable simplicity is known for Wagstaff numbers. A natural Chebyshev/Lucas-type candidate — the single congruence

$$\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}, \qquad \omega_3 = 3 + 2\sqrt{2},$$

in $\mathbb{Z}[\sqrt{2}]$ (the $a = 3$ case of the Chua framework [7]) — is known to hold for every Wagstaff prime, but its sufficiency remains conjectural.

In the absence of a deterministic standalone test, the Brillhart–Lehmer–Selfridge (BLS) framework [3] provides unconditional primality proofs whenever $N \pm 1$ has a "large enough" completely factored divisor: in the $N-1$ form used here, one needs a divisor $F \mid N-1$ with $F^3 > N$ and with every prime of F independently certified. We present BLS-based primality proofs for the three exponents $p \in \{2617, 10501, 12391\}$.

1.2. **Motivation.** The three exponents addressed here are already known prime; the contribution of the present paper is methodological, not existence-of-a-prime. Two points motivate revisiting them via BLS:

- *Auditability.* A BLS certificate is a short JSON record of per-prime witnesses plus a discriminant check; re-verification is a handful of modular exponentiations whose correctness rests on standard Fermat-type arithmetic and on APR-CL for each cofactor. ECPP certificates are authoritative but substantially larger and require elliptic-curve evaluation to replay; a BLS/APR-CL pipeline provides an independent confirmation of the primality claim.
- *Technique demonstration.* The cyclotomic harvesting of factored content from $2^{p-1} - 1 = \prod_{d|p-1} \Phi_d(2)$ for the Wagstaff $N-1$ side is straightforward in principle but, so far as we are aware, has not been applied in the published literature to the specific exponents $p \in \{2617, 10501, 12391\}$. The procedure is generalisable: any Wagstaff exponent p with sufficiently smooth $p-1$ and sufficient Cunningham coverage is a candidate.

1.3. **This paper.** We give three unconditional BLS primality proofs, one per exponent. The proofs and the cross-check rest on three ingredients:

(1) Establishing that Condition (II), $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$, holds for every Wagstaff prime W_p (Proposition 2.3) — this is the $a = 3$ case of the Chua framework [7] and follows from $W_p \equiv 3 \pmod 8$.
(2) For the three target exponents $p \in \{2617, 10501, 12391\}$, applying the BLS $N-1$ criterion (Theorem 3.1) to the cyclotomic decomposition $2^{p-1} - 1 = \prod_{d|p-1} \Phi_d(2)$, harvesting factored portions from the Cunningham project tables [6] plus direct ECM/$p-1$/Pollard-ρ computation. Every cofactor primality is certified by APR-CL [1, 8].
(3) Verifying Condition (II) independently at each of the three W_p as an auxiliary cross-substrate check on the $\mathbb{Z}[\sqrt{2}]$ arithmetic implementation (a distinct code path from the $\mathbb{Z}$-only BLS arithmetic).

The resulting proofs are short (Section 4), fully unconditional, and cleanly separated from the Chebyshev sufficiency conjecture (not used, not needed). All computations are reproducible from the scripts described in Section 5.

1.4. **Notation.** Throughout, $p \geq 5$ denotes an odd prime and $W_p = (2^p + 1)/3$. For an integer or algebraic integer x coprime to a modulus m, $\mathrm{ord}_m(x)$ denotes the multiplicative order of x modulo m; $v_q(\cdot)$ denotes the q-adic valuation. We work in $\mathbb{Z}[\sqrt{2}]$, with fundamental unit $\alpha = 1 + \sqrt{2}$ and Chebyshev base $\omega_3 = \alpha^2 = 3 + 2\sqrt{2}$. The Pell sequences (U_n) and (V_n) are defined by $\alpha^n = V_n/2 + U_n\sqrt{2}$, with the recurrence $X_{n+1} = 2X_n + X_{n-1}$ and the identity $V_n^2 - 8U_n^2 = 4(-1)^n$. The Jacobi symbol is written (a/n).

2. Chebyshev bases and the Chua framework

2.1. **The Chua identity.** For an integer $a \neq 0, \pm 1$, set $\omega_a = a + \sqrt{a^2 - 1}$, so that $\omega_a \bar{\omega}_a = 1$ and $\omega_a^n + \omega_a^{-n} = 2T_n(a)$ where T_n is the n-th Chebyshev polynomial of the first kind — hence the name "Chebyshev base".

Theorem 2.1 (Chua [7]; cf. Williams [11, Ch. 4]). *Let Q be an odd prime with $\gcd(a^2 - 1, Q) = 1$. Set $D = a^2 - 1$, $\varepsilon = (D/Q)$, $\delta = (2(a+1)/Q)$. Then*

$$\omega_a^{(Q-\varepsilon)/2} \equiv \delta \pmod{Q}$$

in $\mathbb{Z}[\sqrt{D}]/(Q)$.

Proof. Define $\gamma = a + 1 + \sqrt{D}$; then $\gamma^2 = 2(a+1) \cdot \omega_a$ in $\mathbb{Z}[\sqrt{D}]$. Two evaluations of γ^Q in $\mathbb{Z}[\sqrt{D}]/(Q)$:

- Frobenius: $\sqrt{D}^Q \equiv \varepsilon\sqrt{D}$ (where $\varepsilon = (D/Q) \in \{\pm 1\}$), so $\gamma^Q \equiv a + 1 + \varepsilon\sqrt{D}$.
- Via the square: $\gamma^Q = \gamma \cdot (\gamma^2)^{(Q-1)/2} = \gamma \cdot (2(a+1))^{(Q-1)/2}\omega_a^{(Q-1)/2} \equiv \gamma \cdot \delta \cdot \omega_a^{(Q-1)/2}$, where $\delta = (2(a+1)/Q)$ by Euler's criterion.

Equating: $\gamma \cdot \delta \cdot \omega_a^{(Q-1)/2} \equiv a + 1 + \varepsilon\sqrt{D}$ in $\mathbb{Z}[\sqrt{D}]/(Q)$.

Case $\varepsilon = +1$. The right-hand side equals γ, and γ is a unit modulo Q (its norm is $(a+1)^2 - D = 2(a+1)$, coprime to Q by $\gcd(D, Q) = \gcd(a^2 - 1, Q) = 1$, which implies $\gcd(2(a+1), Q) = 1$ since Q is odd). Dividing by γ yields $\delta \cdot \omega_a^{(Q-1)/2} \equiv 1$, i.e. $\omega_a^{(Q-1)/2} \equiv \delta$.

Case $\varepsilon = -1$. The right-hand side equals $\bar{\gamma} = a + 1 - \sqrt{D}$, the conjugate of γ. A direct computation gives $\bar{\gamma}/\gamma = \bar{\gamma}^2/(\gamma\bar{\gamma}) = ((a+1) - \sqrt{D})^2/(2(a+1)) = a - \sqrt{D} = \omega_a^{-1}$. So $\delta \cdot \omega_a^{(Q-1)/2} \equiv \omega_a^{-1}$, i.e. $\omega_a^{(Q+1)/2} \equiv \delta$.

The two cases combine as $\omega_a^{(Q-\varepsilon)/2} \equiv \delta \pmod{Q}$. □

2.2. **The base $a = 3$.**

Lemma 2.2 (mod-8 residue). *For every odd prime $p \geq 3$, $W_p \equiv 3 \pmod{8}$.*

Proof. Since p is odd, $3W_p = 2^p + 1 \equiv 1 \pmod{8}$, so $W_p \equiv 3 \pmod{8}$. □

Proposition 2.3 (Base $a = 3$ necessity). *For every Wagstaff prime W_p with $p \geq 5$,*

$$\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}, \qquad \omega_3 = 3 + 2\sqrt{2}.$$

Proof. For $a = 3$: $D = a^2 - 1 = 8$ and $2(a+1) = 8$, so $\varepsilon = \delta = (8/W_p) = (2/W_p)^3 = (2/W_p)$. By Lemma 2.2, $W_p \equiv 3 \pmod 8$, so the second supplement to quadratic reciprocity gives $(2/W_p) = -1$; hence $\varepsilon = \delta = -1$. Theorem 2.1 with $Q = W_p$ yields $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$. □

Remark 2.4 (Role of Condition (II) in this paper)**.** Proposition 2.3 is not used to deduce primality in this paper; it is verified at each of the three target W_p only as an independent check on the $\mathbb{Z}[\sqrt{2}]$ arithmetic used in the supplementary code. Primality is established by Theorem 3.1 alone.

Remark 2.5 (Uniqueness)**.** The base $a = 3$ is the unique integer $a \geq 2$ with $a^2 - 1$ a power of 2. Indeed, $(a-1)(a+1) = 2^k$ with $a-1, a+1$ two consecutive even numbers forces $\gcd(a-1, a+1) = 2$, so both are powers of 2 differing by 2 — hence $\{a-1, a+1\} = \{2, 4\}$ and $a = 3$. Consequently, $a = 3$ is the unique Chua base for which both $\varepsilon = (D/Q)$ and $\delta = (2(a+1)/Q)$ are controlled by the single Jacobi symbol $(2/Q)$ — in particular, determined by Q mod 8 alone. Other bases a give conditions whose truth depends on residues of Q modulo further prime divisors of $a^2 - 1$; only $a = 3$ yields a condition universal across W_p (and more generally across all $Q \equiv 3 \pmod 8$).

3. The BLS $N - 1$ theorem

The primality proofs of Section 4 rely on the classical BLS $N - 1$ theorem, which we recall here.

Theorem 3.1 (BLS $N - 1$, [3, Theorem 5 and Theorem 11])**.** *Let $N > 1$ be an odd integer. Write $N - 1 = FR$ with $F > 1$, $\gcd(F, R) = 1$, and F completely factored into known primes. Suppose for each prime $q \mid F$ there exists an integer a_q with*

$$a_q^{N-1} \equiv 1 \pmod N \qquad \text{and} \qquad \gcd(a_q^{(N-1)/q} - 1,\ N) = 1.$$

If $F > \sqrt{N}$, then N is prime. More generally, if $F^3 > N$ (i.e. $F > N^{1/3}$), write $N = mF + s + 1$ with $0 \leq s < F$ and set $\Delta = s^2 - 8m$; if Δ is not a perfect square (in particular, if $\Delta < 0$), then N is prime.

The $F > N^{1/3}$ threshold with the discriminant check is BLS's central improvement over the classical $F > N^{1/2}$: instead of needing half the bit-size of N to be factored, we need only a third. The quantity reported in the margin column of Table 1 is $M := \lfloor \log_2 F^3 \rfloor - \lfloor \log_2 N \rfloor$, the integer bit-count by which F^3 exceeds N in bit-length; $M \geq 1$ already implies $F^3 > N$. For the three exponents below, $M \in \{46, 3261, 2860\}$, so the threshold is met with considerable slack.

4. The three primality proofs

4.1. **Framework.** The three proofs in §§4.2–4.4 invoke Theorem 3.1 (BLS $N - 1$) with specific factorisations of $N - 1 = 2(2^{p-1} - 1)/3$ drawn from the cyclotomic decomposition

$$2^{p-1} - 1 = \prod_{d \mid p-1} \Phi_d(2),$$

harvested partly from the Cunningham project tables [6] and partly from direct computation. Each factor of each $\Phi_d(2)$ contributes to the BLS factored portion F once its primality is certified by APR-CL [1, 8]; Fermat-type witnesses $a \in \{2, 3, 5, \dots\}$ are then validated for each prime $q \mid F$. The exact software toolchain and version numbers used to produce the certificates are documented in Section 5. The proofs are unconditional and do not rely on the Chebyshev sufficiency conjecture.

At each p, we additionally verify the Chua $N+1$ condition $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ (Proposition 2.3) as an independent implementation check: BLS uses only $\mathbb{Z}$ arithmetic, whereas Condition (II) exercises $\mathbb{Z}[\sqrt{2}]$ ring multiplication, a distinct code path in the supplementary software. The Chua congruence is recorded only as an auxiliary check; primality is established solely via Theorem 3.1 (see Remark 2.4).

APR-CL certification is applied uniformly to every prime q of F, regardless of size — the smallest single-digit primes receive the same unconditional APR-CL treatment as the largest multi-hundred-digit factors.

p	W_p digits	$\tau(p-1)$	primes in F	M (bits)	wall time
2617	788	16	22	46	< 10 s
10501	3161	48	103	3261	98 s
12391	3730	32	61	2860	82 s

TABLE 1. Summary of the three BLS $N-1$ proofs. $\tau(p-1)$ is the number of divisors of $p-1$; “primes in F” is the number of distinct primes in the factored portion; wall times are single-core on commodity hardware, see Section 5. Full per-cofactor data is in the JSON certificates.

Full per-proof data is archived in `bls_certificate_w{2617,10501,12391}.json` (Section 5).

4.2. W_{2617}.

Theorem 4.1. *The Wagstaff number W_{2617} is prime.*

Proof. Set $N = W_{2617}$, a 788-digit number. We apply Theorem 3.1 to the $N-1$ side; the factored portion F will be shown to exceed $N^{1/3}$.

The exponent satisfies $p - 1 = 2616 = 2^3 \cdot 3 \cdot 109$, with 16 divisors. The cyclotomic decomposition

$$2^{p-1} - 1 \;=\; \prod_{d \mid p-1} \Phi_d(2)$$

gives 16 factors, of which 12 (those with $d \le 654$) are fully factored by direct computation. Their prime factors, together with the external factor 2 of $N - 1 = 2(2^{p-1} - 1)/3$, constitute the BLS factored portion F; it contains 22 distinct primes and satisfies $M = 46$ (Table 1), well above the $F > N^{1/3}$ threshold.

For each prime $q \mid F$ (the 22 primes constituting the factored portion), the witness $a_q \in \{2, 3\}$ is tested: $a_q^{N-1} \equiv 1 \pmod N$ and $\gcd(a_q^{(N-1)/q} - 1, N) =$

1. Every witness succeeds (full data in `bls_certificate_w2617.json`; digest `823f...c719` in Appendix A). Writing $N = mF + s + 1$ with $0 \le s < F$ and $\Delta = s^2 - 8\,m$, the discriminant Δ is not a perfect square.

By Theorem 3.1, $N = W_{2617}$ is prime. □

4.3. W_{10501}.

Theorem 4.2. *The Wagstaff number W_{10501} is prime.*

Proof. Set $N = W_{10501}$, a 3161-digit number. The exponent $p-1 = 10500 = 2^2 \cdot 3 \cdot 5^3 \cdot 7$ has 48 divisors. Of the 48 cyclotomic factors $\Phi_d(2)$ of $2^{p-1}-1$, 43 are fully factored — 38 by direct computation (using SymPy `factorint` with Pollard ρ, $p-1$, and ECM) and 5 retrieved from the Cunningham project tables [6]. This contributes 103 primes to the BLS factored portion F, each certified prime by APR-CL [1, 8].

The factored portion satisfies $F^3 > N$ by a margin of 3261 bits, substantially exceeding the BLS threshold. Fermat-type witnesses $a_q \in \{2, 3, 5\}$ are validated for each of the 103 primes $q \mid F$; every witness succeeds. The discriminant $\Delta = s^2 - 8\,m$ (with $N = mF + s + 1$, $0 \le s < F$) is computed and verified non-square. Full data in `bls_certificate_w10501.json`; digest `c7b3...9ea5` in Appendix A.

By Theorem 3.1, $N = W_{10501}$ is prime. □

4.4. W_{12391}.

Theorem 4.3. *The Wagstaff number W_{12391} is prime.*

Proof. Set $N = W_{12391}$, a 3730-digit number. The exponent $p-1 = 12390 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 59$ has 32 divisors. Of the 32 cyclotomic factors $\Phi_d(2)$, 27 are fully factored: 19 by direct computation, 5 retrieved from the Cunningham tables [6], 2 retrieved from FactorDB [9] (used as a discovery aid only — every such factor is re-certified by APR-CL [1, 8] before entering F), and 1 itself prime. This contributes 61 primes to F, the largest being a 371-digit prime cofactor of $\Phi_{2065}(2)$. Each of the 61 primes receives independent APR-CL certification.

The factored portion F satisfies $F^3 > N$ by a 2860-bit margin. Fermat-type witnesses $a_q \in \{2, 3\}$ are validated for each of the 61 primes; every witness succeeds. The discriminant $\Delta = s^2 - 8\,m$ (with $N = mF + s + 1$, $0 \le s < F$) is non-square. Full data in `bls_certificate_w12391.json`; digest `1b0d...3570` in Appendix A.

By Theorem 3.1, $N = W_{12391}$ is prime. □

4.5. **Limits of the method.**

Remark 4.4 (BLS ceiling)**.** The BLS $N-1$ approach requires $p-1$ to be sufficiently smooth that enough cyclotomic factors $\Phi_d(2)$ with $d \mid p-1$ have known factorisations. Table 2 tabulates $p-1$ for every Wagstaff prime and probable prime with $p > 2617$ through the present computational horizon $p \le 267017$. A divisor $d \mid p-1$ contributes to F only if $\Phi_d(2)$ is completely factored (in Cunningham tables, FactorDB, or by direct computation). For most of the exponents listed, $p-1$ has a large prime factor ℓ, forcing the divisor $d = \ell$ (or $d = 2\ell$) to dominate $2^{p-1}-1$, and no general-purpose factorisation of the corresponding $\Phi_d(2)$ or $\Phi_{2\ell}(2)$ is known at these sizes.

For the factorisation data consulted here, only $p = 10501$ and $p = 12391$ yield a factored portion large enough for the present BLS $N-1$ approach; $p = 14479$ is the closest near-miss.

p	$p-1$ factorisation	BLS $N-1$ feasible?
3539	$2 \cdot 29 \cdot 61$	no: $\Phi_{1769}(2) \sim 2^{1680}$ unfactored
5807	$2 \cdot 2903$ (2903 prime)	no: $\Phi_{2903}(2) \sim 2^{2902}$ unfactored
10501	$2^2 \cdot 3 \cdot 5^3 \cdot 7$	**yes** (proved, §4.3)
10691	$2 \cdot 5 \cdot 1069$ (1069 prime)	no: $\Phi_{1069}(2) \sim 2^{1068}$ unfactored
11279	$2 \cdot 5639$ (5639 prime)	no: $\Phi_{5639}(2) \sim 2^{5638}$ unfactored
12391	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 59$	**yes** (proved, §4.4)
14479	$2 \cdot 3 \cdot 19 \cdot 127$	no: $\Phi_{14478}(2)$ only partially factored
42737	$2^4 \cdot 2671$ (2671 prime)	no: $\Phi_{2671}(2) \sim 2^{2670}$ unfactored
83339	$2 \cdot 41669$ (prime)	no (PRP; large prime in $p-1$)
95369	$2^3 \cdot 7 \cdot 1703$	no (PRP; $\Phi_{1703}(2)$ unfactored)
117239	$2 \cdot 58619$ (prime)	no (PRP; large prime in $p-1$)
127031	$2 \cdot 5 \cdot 12703$ (prime)	no (PRP; large prime in $p-1$)
138937	$2^3 \cdot 3 \cdot 7 \cdot 827$	no (PRP; $\Phi_{827}(2)$ partial)
141079	$2 \cdot 3 \cdot 7 \cdot 3359$ (prime)	no (PRP; large prime in $p-1$)
267017	$2^3 \cdot 33377$ (prime)	no (PRP; large prime in $p-1$)

Table 2. Smoothness of $p-1$ for every Wagstaff prime or probable prime with $2617 < p \leq 267017$. The eight proved Wagstaff primes above 2617 are listed first, followed by the seven probable primes above the current proving horizon. A detailed computation of the bit contribution from each divisor $d \mid p-1$ is recorded in the JSON certificates for the two feasible cases; for the others, the listed large prime factor of $p-1$ forces a $\Phi_d(2)$ whose full factorisation is currently out of reach.

Extensions of the Cunningham tables [6] — or direct ECM on the divisors $\Phi_d(2)$ for $d \mid p-1$ with these target exponents — would push the feasibility frontier. The cofactor data recorded in the JSON certificates (Section 5) identifies exactly which $\Phi_d(2)$ remain only partially factored for $p = 10501$ and $p = 12391$; these are natural targets for further ECM work.

5. Computational details

5.1. **Software toolchain.** All runs reported in this paper used Python 3.12.3, SymPy 1.14.0, PARI/GP 2.15.4 (linked against GMP 6.3.0), and gmpy2 2.3.0 on Ubuntu 24.04 LTS (AMD Ryzen 5 3600, 64 GB RAM); the scripts in the supplementary repository are compatible with Python ≥ 3.10, SymPy ≥ 1.12, PARI/GP ≥ 2.15, and gmpy2 ≥ 2.1, and use only free, open-source software.

- *Integer factorisation* of cyclotomic factors $\Phi_d(2)$: SymPy's `factorint` (combining trial division, Pollard ρ, Pollard $p-1$, and ECM internally), augmented for cyclotomic factors by trial division along the arithmetic progression $r \equiv 1 \pmod d$ implied by primitive-divisor theory.

- *Pre-known large factorisations.* The Cunningham project tables [6] (accessed January–April 2026) supplied factorisations for cyclotomic factors $\Phi_d(2)$ with $d \leq 1200$, and FactorDB [9] was used as a discovery aid for two larger terms ($\Phi_{2065}(2)$ and $\Phi_{2478}(2)$ in Theorem 4.3). Every retrieved factor is independently re-certified for primality by APR-CL before entering F, and the driver scripts re-check the product $\prod_i p_i^{e_i} = \Phi_d(2)$ exactly before accepting a factorisation; FactorDB plays no evidentiary role.
- *Primality of cofactors* (every prime used in any BLS certificate): PARI/GP `isprime(n, 2)`, which forces the APR-CL (Adleman–Pomerance–Rumely / Cohen–Lenstra) path [1, 8], implemented in PARI/GP along the lines of Bosma and van der Hulst [4], and is deterministic. The largest cofactor is a 371-digit prime factor of $\Phi_{2065}(2)$ (Theorem 4.3); the smallest are single-digit primes, but all receive APR-CL certification for uniformity.
- *Fast modular arithmetic* for the exponentiations $a^{N-1} \bmod N$ and $\omega_3^{(N+1)/2} \bmod N$ at each of the three W_p: Python's built-in big-int `pow`, with GMP-backed acceleration via gmpy2 when installed.

Remark 5.1 (Factor provenance)**.** Each JSON certificate records a `factor_provenance` field: every prime $q \mid F$ is tagged with the source from which its literal value first entered the driver. The labels used are `algebraic` (the prime 2 contributed by $N-1 = 2\,(2^{p-1}-1)/3$), `cunningham` (factor drawn from a Cunningham project table literal), `factordb` (obtained via FactorDB lookup), `direct_sympy` (produced by `sympy.factorint` on $\Phi_d(2)$), `trial_div_cyclotomic` (discovered by trial division along the progression $r \equiv 1 \pmod{d}$), `cyclotomic_prime` ($\Phi_d(2)$ itself prime), and `residual_prime_aprcl` (residual cofactor of $\Phi_d(2)$ after removing known factors, verified prime before entering F). Provenance is audit metadata only: every prime, regardless of label, receives the same APR-CL certification before being admitted to F.

The driver scripts `bls_n_minus_1_w{2617,10501,12391}.py` in the supplementary repository emit JSON certificates (`bls_certificate_w*.json`) recording every cyclotomic factor, prime decomposition, BLS witness, and primality method used, along with full execution logs. A master verifier `verify_bls.py` re-reads each certificate from disk, treats every claim in it as untrusted input, rebuilds F from $v_q(N-1)$, re-runs APR-CL on every prime, re-checks the witnesses and the discriminant, and re-runs Condition (II); any disagreement exits non-zero.

5.2. **Supplementary material.** All data and scripts are archived on Zenodo. The concept DOI https://doi.org/10.5281/zenodo.19643792 always resolves to the latest version; the version-specific DOI for the release accompanying this paper is https://doi.org/10.5281/zenodo.19645441. A development mirror is hosted on GitHub at https://github.com/dolonet/wagstaff-bls-primality; the GitHub repository is a mirror, and the Zenodo deposit is the citable artifact.

The archive contains, for each $p \in \{2617, 10501, 12391\}$:

(1) the BLS primality certificate (JSON), recording every cyclotomic factor $\Phi_d(2)$, its prime decomposition, the BLS witness a for each prime $q \mid F$, and the APR-CL timings for each cofactor;
(2) the driver script `bls_n_minus_1_w<p>.py` reproducing the certificate end-to-end;
(3) the full APR-CL primality log for each cofactor (each prime q is re-checked by piping the PARI/GP statement `isprime(<q>, 2)` to `gp -q`);
(4) the independent Condition (II) verification log.

5.3. **Verification protocol.** A referee can verify each primality claim from scratch with the following four-step protocol:

(1) *Fetch.* Download the Zenodo archive at the version-specific DOI for this release (see Section 5.2) and extract.
(2) *Run the master verifier.* For each $p \in \{2617, 10501, 12391\}$, execute `python3 scripts/verify_bls.py data/bls_certificate_w<p>.json`. `verify_bls.py` treats every claim in the certificate as untrusted input: it re-reads N from the exponent, independently rebuilds F from the recorded $v_q(N-1)$ data, re-runs PARI/GP `isprime(q, 2)` on every prime in the decomposition, re-checks the Fermat witnesses, verifies $F^3 > N$ and $\gcd(F, R) = 1$, re-computes and re-tests the BLS discriminant, and re-runs Condition (II) in $\mathbb{Z}[\sqrt{2}]/(N)$. Any discrepancy exits non-zero.
(3) *Re-verify the factorisation.* For each recorded factor, the driver re-runs APR-CL primality and re-computes the product $\prod_i p_i^{e_i} = \Phi_d(2)$ exactly. Small $\Phi_d(2)$ are additionally re-factored from scratch via SymPy; for those $\Phi_d(2)$ whose full factorisation relies on external sources (Cunningham tables and FactorDB, tagged as such in the `factor_provenance` field of Remark 5.1), the recorded factorisation is treated as input to be verified, not regenerated — their large primes (e.g. the 371-digit prime factor of $\Phi_{2065}(2)$) cannot be rediscovered by SymPy in any realistic time.
(4) *Inspect logs.* The `logs/` directory contains the full stdout of the runs that produced the certificates on the original hardware, for line-by-line comparison.

The archive is deterministic and version-pinned; running the drivers and verifier on a clean machine reproduces every numeric result in this paper.

6. Summary

This paper delivers three unconditional primality proofs of W_{2617}, W_{10501}, and W_{12391} via the BLS $N-1$ criterion (Theorems 4.1, 4.2, 4.3), independent of ECPP. Every prime of every factored portion is certified by APR-CL; each certificate also records, as an independent implementation check, the Chua $a = 3$ congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ in $\mathbb{Z}[\sqrt{2}]/(W_p)$. All primality certificates, driver scripts, and full run logs are archived on Zenodo under a concept DOI.

Table 2 makes the reach of the present technique concrete: within the 36 known Wagstaff primes and probable primes, the only two exponents above

2617 for which $p-1$ is smooth enough to meet the BLS factored-portion threshold on current Cunningham and FactorDB data are $p = 10501$ and $p = 12391$ — precisely the two targets of this paper. The closest near-miss is $p = 14479$, where the relevant cyclotomic factor $\Phi_{14478}(2)$ is only partially factored; the remaining Wagstaff primes and probable primes in the interval $2617 < p \leq 267017$ are out of reach of BLS $N-1$ at present because $p-1$ contains a large prime factor forcing an unfactored $\Phi_d(2)$.

Appendix A. Certificate digests

For each $p \in \{2617, 10501, 12391\}$, Table 3 records a SHA-256 digest of the corresponding JSON certificate, together with the key aggregate data extracted from it. The digest is computed over the canonical serialisation `json.dumps(obj, sort_keys=True, separators=(',', ':'))`, so any byte-level modification of the certificate changes the recorded hash. A referee running `verify_bls.py` can reproduce each digest from the archived JSON before accepting the file as the input of the re-verification protocol.

p	F digits	primes in F	M (bits)	largest q	SHA-256 (first 16 hex)
2617	268	22	46	50 d	`823fad79bef1f5eb`
10501	1381	103	3261	160 d	`c7b3d40558d723e2`
12391	1531	61	2860	371 d	`1b0dc924417d9fa7`

Table 3. Per-proof certificate digests and aggregates. "largest q" is the digit count of the largest prime in the factored portion F; every such prime receives an unconditional APR-CL certificate. Full SHA-256 hashes: `823fad79bef1f5eb27218245068da27ee0790991bb28b328482ddda93cf1c719` (W_{2617}), `c7b3d40558d723e2818a147f9f7cfca11ae58601bbf0e7dceb1bb5d442c39ea5` (W_{10501}), `1b0dc924417d9fa7411c1bc830b7114adaac3782c0c10e82938dce848f063570` (W_{12391}).

The archived certificates also record: (i) the complete prime factorisation of F with provenance tags; (ii) the Fermat-type witness a_q for each prime $q \mid F$; (iii) the APR-CL wall time for each such prime; and (iv) the BLS discriminant $\Delta = s^2 - 8\,m$, written as a large integer in decimal, with the square-test outcome.

Appendix B. Reference data

B.1. **The 36 known Wagstaff primes and probable primes.** Proved prime ($p \leq 42737$, by ECPP [2] or by this paper), excluding the trivial case $p = 3$ for which $W_3 = 3$:

$$p \in \{5, 7, 11, 13, 17, 19, 23, 31, 43, 61, 79, 101, 127, 167, 191, 199, 313, 347, 701, 1709, 2617, 3539, 5807, 10501, 10691, 11279, 12391, 14479, 42737\}$$

— 29 exponents. Probable prime ($p > 42737$, not yet proved but consistent with every applied pseudoprime test):

$$p \in \{83339, 95369, 117239, 127031, 138937, 141079, 267017\}$$

— 7 exponents. The "36 known" count in the header includes both groups.

B.2. **Cyclotomic decomposition of $W_p - 1$.** For odd prime p and $N = W_p$:

$$N - 1 = \frac{2^p - 2}{3} = \frac{2(2^{p-1} - 1)}{3}.$$

The cyclotomic decomposition

$$2^{p-1} - 1 = \prod_{d \mid p-1} \Phi_d(2)$$

harvests factored portions from divisors d of $p - 1$; each $\Phi_d(2)$ is a positive integer whose prime factors lie in an arithmetic progression $r \equiv 1 \pmod{d}$. This is the source of the BLS factored portions in §§4.2–4.4.

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